Homework 2: Cosmology

Solving the Friedmann Equation

The Friedmann Equation for a matter dominated universe in non-dimensionalized form is

$$(\tilde{R}')^2 - 2\tilde{R} = -k\tilde{R}^2 \tag{1}$$

where $\tilde{R} = R/R^*$, and the prime denotes derivation with respect to the non-conformal time $d\eta = \frac{c\,dt}{R}$.

1. Solve the matter-dominated Friedmann Equation for all three cases (closed, flat, open) to obtain an expression for $R(\eta)$ and $t(\eta)$.

   If $k = 0$ and $\tilde{R}(0) = 0$, then we obtain:

$$\sqrt{(\tilde{R}')^2} = \sqrt{2\tilde{R}} \qquad \text{by addition}$$

$$\frac{1}{\sqrt{2}}\frac{\tilde{R}^{1/2}}{1/2} = \int \frac{d\tilde{R}}{\sqrt{2\tilde{R}}} = \int d\eta = \eta \qquad \text{by integration}$$

$$\tilde{R} = \frac{\eta^2}{2}$$

   If $k = \pm 1$ and $\tilde{R}(0) = 0$, then we obtain:

$$\sqrt{(\tilde{R}')^2} = \sqrt{\frac{1}{k} - k\tilde{R}^2 + 2\tilde{R} - \frac{1}{k}}$$

$$\frac{\sqrt{k}}{k}\int \frac{k\ d\tilde{R}}{\sqrt{1-(k\tilde{R}-1)^2}} = \int 1\ d\eta$$

$$\frac{1}{\sqrt{k}}\sin^{-1}(k\tilde{R}-1) = \eta + C$$

$$\frac{R}{R_0} = \tilde{R}(\eta) = \frac{\sin\left(\frac{k\eta}{\sqrt{k}}\right)+1}{k} = k\sin(\eta\sqrt{k})$$

$$R = \sqrt{\frac{3c^3}{4G\pi\rho\tilde{R}}}$$

   If $k = 1$, then $\tilde{R}(\eta) = 2\sin^2(\eta/2)$. If $k = -1$, then $\tilde{R}(\eta) = 2\sinh^2(\eta/2)$.

   If $\eta(0) = 0$, then we obtain

$$t(\eta) = \int dt = \int \frac{R(\eta)\ d\eta}{c}.$$

2. Use your favorite plotting program to plot the non-dimensionalized scale factor R/R^* versus non-dimensionalized time $ct/(2\pi R^*)$.

Numerically, we will actually solve

$$\frac{c\ dt}{R^*} = \frac{R}{R^*} = \tilde{R}$$

in order to obtain an expression of $\frac{c\ t}{R^*}$ in terms of η, which we then scale by a factor of $\frac{1}{2\pi}$.
See Jupyter notebook attached.
The general Friedmann equation can be solved numerically with a variety of assumptions about matter, radiation, and vacuum density.

3. Show that the Friedmann Equation can be written in non-dimensionalized form: $\left(\frac{da}{d\tilde{t}}\right)^2 = \Omega_m a^{-1} + \Omega_r a^{-2} + \Omega_v a^2 - (\Omega_0 - 1)$, where $\tilde{t} = H_0 t$ and $a = R/R_0$. We obtain the result as follows:

$$\Omega_v + \Omega_m a^{-3} + \Omega_r a^{-4} - (\Omega_0 - 1)a^{-2} = \left(\frac{H}{H_0}\right)^2 = \left(\frac{1/R_0}{RH_0/R_0}\frac{dR}{dt}\right)^2$$

$$a^{-2}\left(\Omega_m a^{-1} + \Omega_v a^2 + \Omega_r a^{-2} - (\Omega_0 - 1)a^0\right) = \left(\frac{1}{R/R_0}\frac{d(R/R_0)}{d(H_0 t)}\right)^2$$

$$\left(\frac{1}{a}\right)^2\left(\Omega_m a^{-1} + \Omega_r a^{-2} + \Omega_v a^2 - (\Omega_0 - 1)a^0\right) = \left(\frac{1}{a}\right)^2\left(\frac{da}{d\tilde{t}}\right)^2$$

$$\Omega_m a^{-1} + \Omega_r a^{-2} + \Omega_v a^2 - (\Omega_0 - 1) = \left(\frac{da}{d\tilde{t}}\right)^2$$

4. Use an ODE solver to solve this equation numerically. Choose intial conditions of $a(0) = 0.1$ to avoid an initial singularity. Plot solutions for a flat universe ($\Omega_0 = 1$), with $\Omega_r = 0.4$, $\Omega_m = 0.4$, and $\Omega_v = 0.2$. Plot the results and identify periods of a radiation-dominated, matter-dominated, and vacuum-dominated universe.

   When $0 \leq t \ll 0.7$, we have $a(t) \approx t$ and thus $a^2 < a < a^{-1} < a^{-2}$ implies that the universe is radiation-dominated. At $t = 0.7$, there is a rough equilibrium between radiation and matter. When $0.7 \leq t \leq \sqrt[3]{2} \approx 1.25$, the universe is matter-dominated. When $1.25 \leq t$, the universe is vacuum-dominated.

   Indeed, our figures show that the $a(t)$ curve looks like $t^{1/2}$, $t^{2/3}$, and e^t on those three sections.